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EX NO: 7

Analog Beamforming Analysis Using MATLAB

Aim

To analyze the performance of analog beamforming by studying beamforming gain, beam width,

and main lobe direction using MATLAB.

Software Required

- MATLAB

MATLAB Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
gain = 10*log10(antennas); % Beamforming Gain (dB)
```

```
figure
```

```
bar(antennas, gain)
```

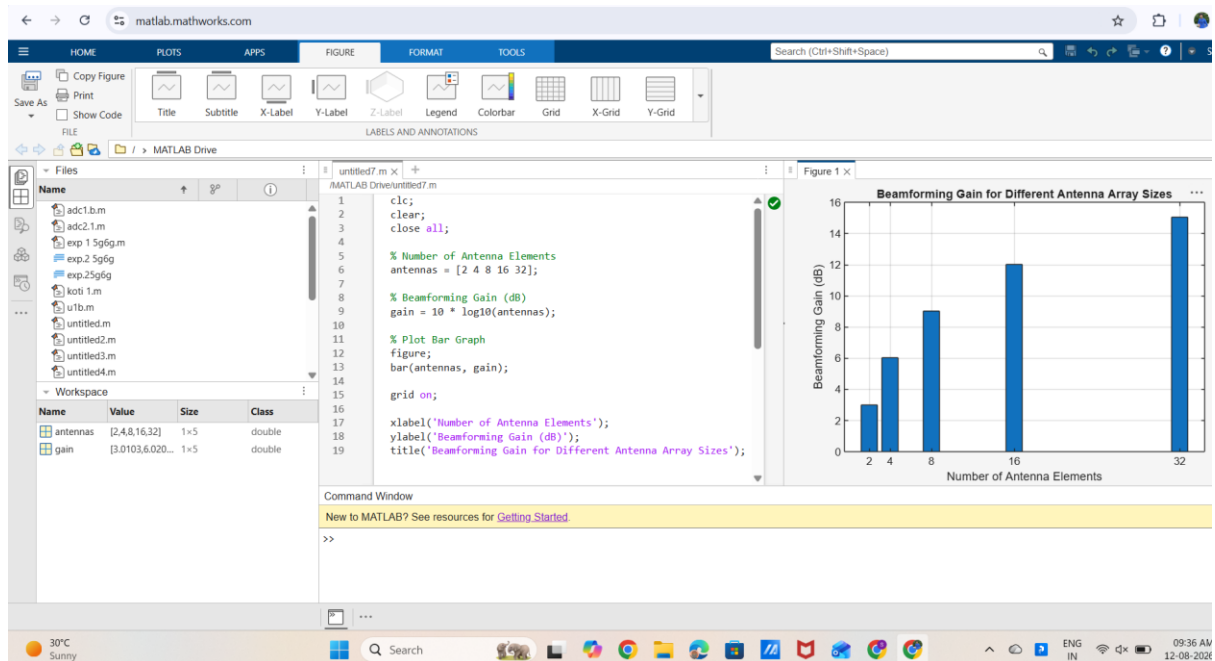
```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beamforming Gain (dB)')
```

```
title('Beamforming Gain for Different Antenna Array Sizes')
```

OUTPUT:



Result:

The MATLAB analysis successfully demonstrated the characteristics of analog beamforming. The results showed that increasing the number of antenna elements improves beamforming gain and reduces beam width, while phase adjustment enables the main lobe to be steered toward different directions.